

DBMS Unit 3 - Normalization Quick Notes

1NF: atomic values, no repeating groups.

2NF: 1NF + no partial dependency on part of a composite key.

3NF: 2NF + no transitive dependency [non-key $\rightarrow$ non-key].

BCNF: for every FD $X \rightarrow Y$, X must be a superkey.

Exam tips [from PYQs]:

- Given $R[A,B,C,D]$ and FDs, find candidate keys, then decompose to BCNF.
- Check lossless join: common attribute must be a key in one relation.
- Dependency preservation: union of projected FDs must imply original FDs.

Shared on StudyVault by Ishan Raj, CSE 2025